

Week 5: Integration of Real life data

Amendments:

Normalization: After integrating the real-world datasets, the database was checked for compliance with First, Second and Third Normal Form. No 1NF, 2NF or 3NF violations were identified. The existing relational structure separates entities such as products, categories, suppliers, supermarkets and disposal methods into their respective tables. Foreign keys are used to represent relationships between these entities, preventing unnecessary duplication of descriptive information.

Real-world Integration: Real-world food-waste datasets were incorporated from Brazil, the United States and Sweden. The data was inserted into the existing relational structure using the product, supermarket and food_waste tables. Because some external datasets did not provide attributes such as price, shelf life, supplier, disposal method or waste reason, these attributes were represented as NULL where appropriate.

Limitations: The integration of real-world data revealed several limitations. The datasets differ in their measurement periods and aggregation levels. For example, some Brazilian data represents individual supermarkets over weekly periods, whereas the US data represents a national annual total. Therefore, directly comparing total quantities between these datasets would not produce a meaningful comparison. In addition, some datasets do not provide waste reasons or disposal methods, limiting the analytical queries that can be performed.